

HW13 Solutions

ECE 310: Digital Signals Processing, Spring 2024

Version: 1.0

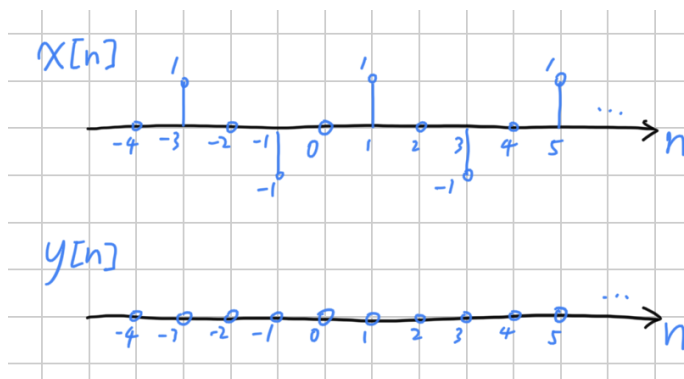
Due Date: 05/01

Prepared by:

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Solution:

- (a) $x[n] = \sin(\frac{\pi}{2}n) = \{\dots, -1, 0, 1, 0, \dots\}$, and $y[n] = x[2n] = \sin(\pi n) = 0$.



- (b)

$$\begin{aligned} X_d(\omega) &= \frac{1}{1 - \frac{1}{2}e^{-j\omega}} \\ Y_d(\omega) &= \frac{1}{2} (X_d(\omega/2) + X_d(\omega/2 - \pi)) \\ &= \frac{1}{2} \left(\frac{1}{1 - \frac{1}{2}e^{-j\omega/2}} + \frac{1}{1 - \frac{1}{2}e^{-j\omega/2}e^{-j\pi}} \right) \\ &= \frac{1}{2} \left(\frac{1}{1 - \frac{1}{2}e^{-j\omega/2}} + \frac{1}{1 + \frac{1}{2}e^{-j\omega/2}} \right) \\ &= \frac{1}{2} \frac{1 - \frac{1}{2}e^{-j\omega/2} + 1 + \frac{1}{2}e^{-j\omega/2}}{(1 - \frac{1}{2}e^{-j\omega/2})(1 + \frac{1}{2}e^{-j\omega/2})} \\ &= \frac{1}{1 - (\frac{1}{2}e^{-j\omega/2})^2} \\ &= \frac{1}{1 - \frac{1}{4}e^{-j\omega}} \end{aligned}$$

Grading: 15 points

- (a) +4 for correct expression of $y[n]$, +4 for correct sketch
- (b) +7 for correct proof

Solution:

(a) By definition, $x[n]$ is determined as

$$x[n] = \{0, 1, 2, 3, 4, 0, 0, 0\}$$

Then by up-sampling, the output is

$$y[n] = \begin{cases} x\left[\frac{n}{3}\right], & n \bmod 3 = 0 \\ 0, & \text{otherwise} \end{cases} = \{0, 0, 0, 1, 0, 0, 2, 0, 0, 3, 0, 0, 4, 0, 0\}.$$

The sketch of $x[n]$ and $y[n]$ is given in Figure 1.

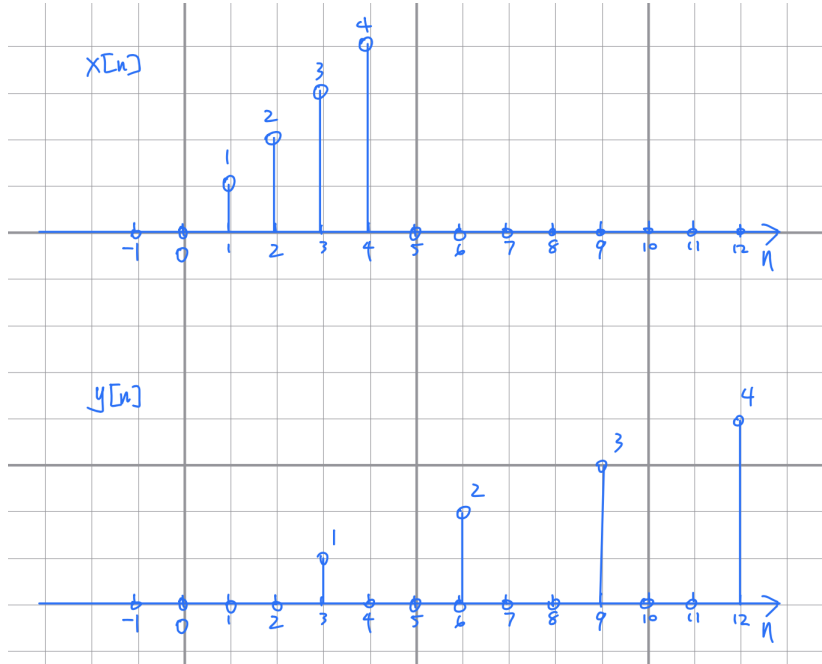


Figure 1: Problem 2(a): The sketch of $x[n]$ and $y[n]$.

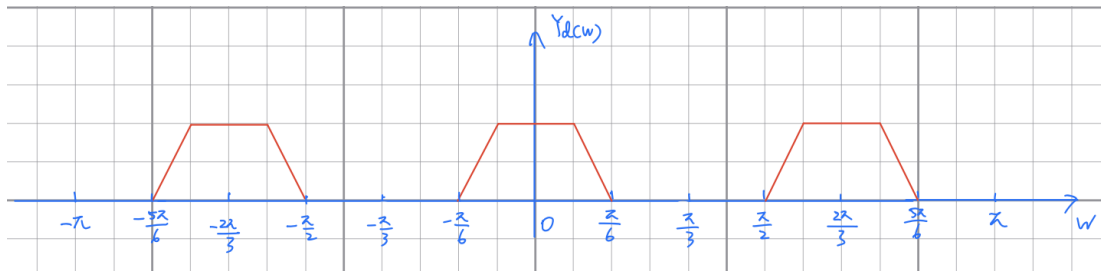


Figure 2: Problem 2(b): The sketch of $Y_d(\omega)$ for $\omega \in [-\pi, \pi]$.

(b) In this case, the relation between $X_d(\omega)$ and its up-sampling version $Y_d(\omega)$ is given as

$$Y_d(\omega) = X_d(3\omega).$$

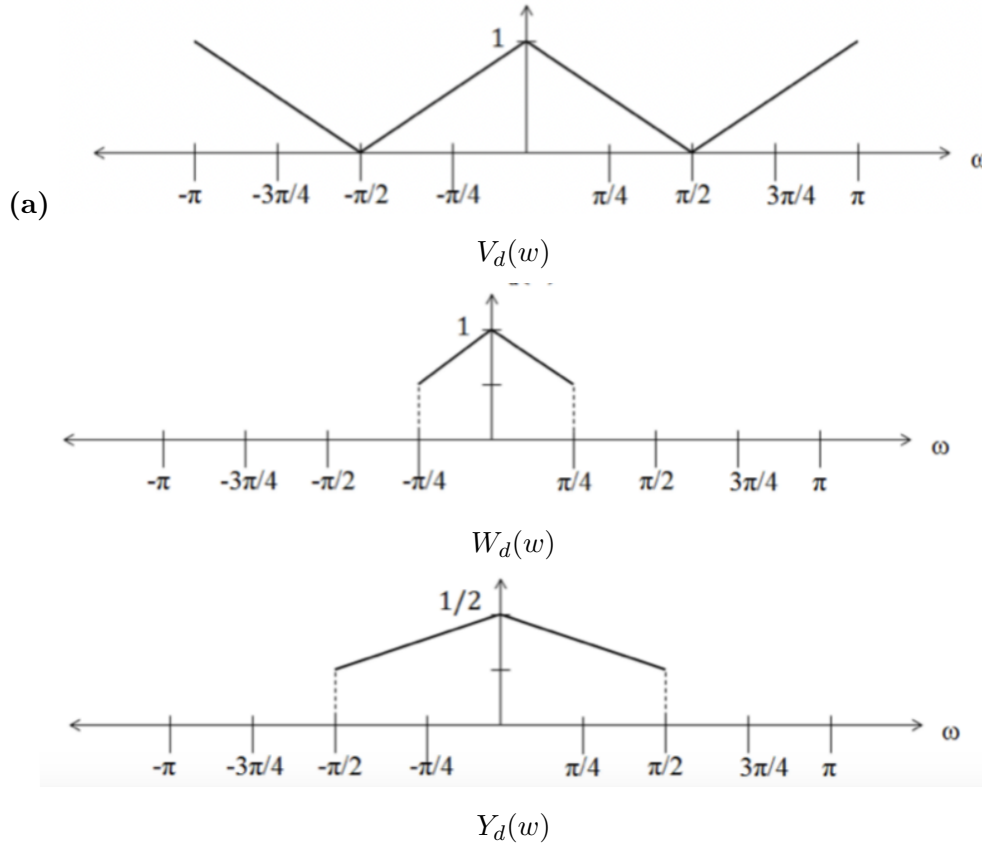
So $Y_d(\omega)$ is a shrinkage of $X_d(\omega)$ by 3. Therefore, for the region $[-\pi, \pi]$ in $Y_d(\omega)$, it represents the region $[-3\pi, 3\pi]$ in $X_d(\omega)$. Based on these, the sketch is given in Figure 2.

Grading: 25 points

- + 15 correct expressions and sketches of $x[n]$ and $y[n]$.
- + 10 correct $Y_d(\omega)$

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Solution:



(b) As we can see from the graphs in a), the whole system behaves like a rectangular low-pass filter that scales the original signal by one-half. So it can be expressed as:

$$H_d(w) = \begin{cases} \frac{1}{2} & 0 < w \leq \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < w \leq \pi \end{cases} \quad (1)$$

or

$$H_d(w) = \frac{1}{2} \text{rect}\left(\frac{w}{\pi}\right)$$

(c) Yes, this is LSI. Since the downsampling module has a factor of 2, the frequency components larger than $\frac{\pi}{2}$ in $W_d(w)$ are at risk of producing aliasing. However, the low pass filter before the downsampling has a cut-off frequency of $\frac{\pi}{4}$ and thus prevents any parts from producing aliasing.

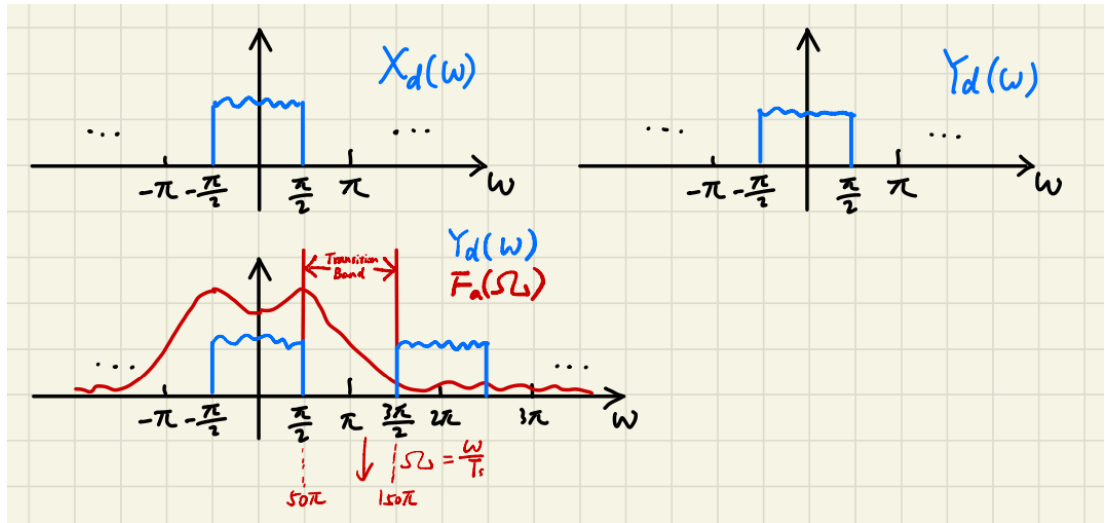
Grading: 25 points

- + 15 correct plots for $V_d(w)$, $W_d(w)$ and $Y_d(w)$.
- + 5 correct closed expressions of the entire system's frequency response.
- + 5 correct statements.

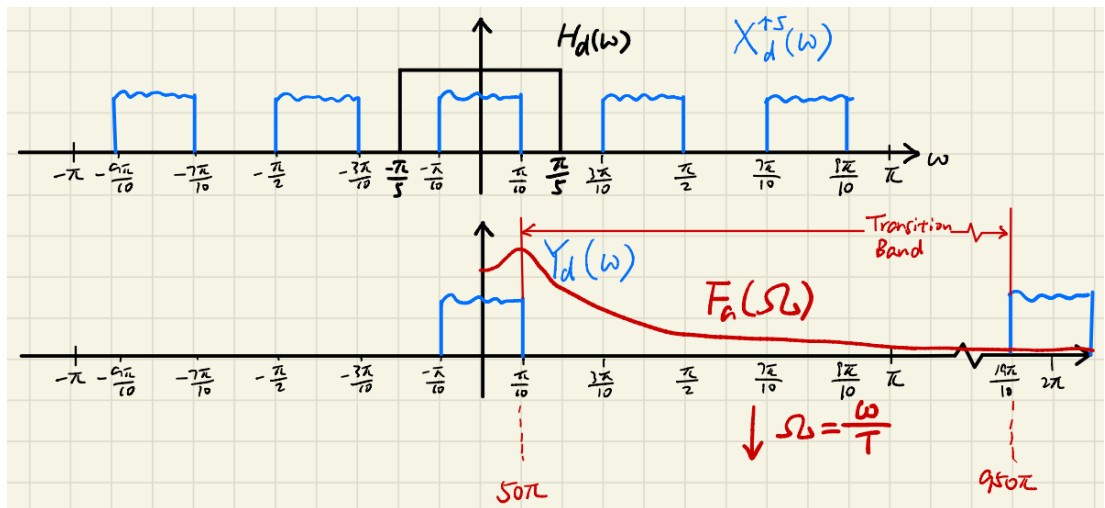
4

Solution:

- (a) The transition band begins from 50π and ends at 150π .



- (b) The transition band begins from 50π and ends at 950π .

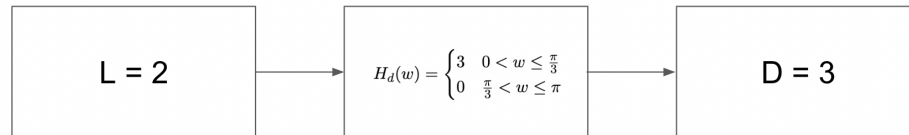


Grading: 15 points

- +7 (a), +8 (b)

5

Solution:



Make sure to upsample before downsampling to avoid aliasing. If the scaling from the sampling is considered, the magnitude of $H_d(w)$ will be 2. Both answers are acceptable.

Grading: 20 points

- +5 each for correct parameters of up-sampler, filter and down-sampler
- +5 if the order of blocks is correct (+2 if up-sampler and down-sampler are flipped)